

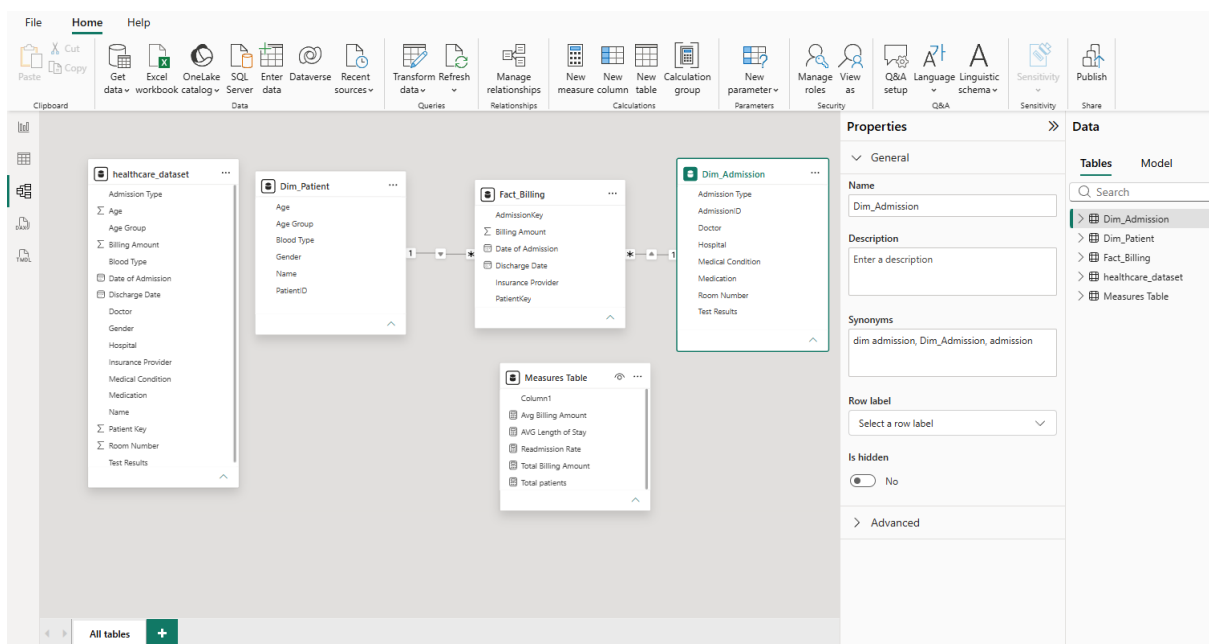
WEEK 6

Healthcare Dashboard

1. Measures Table

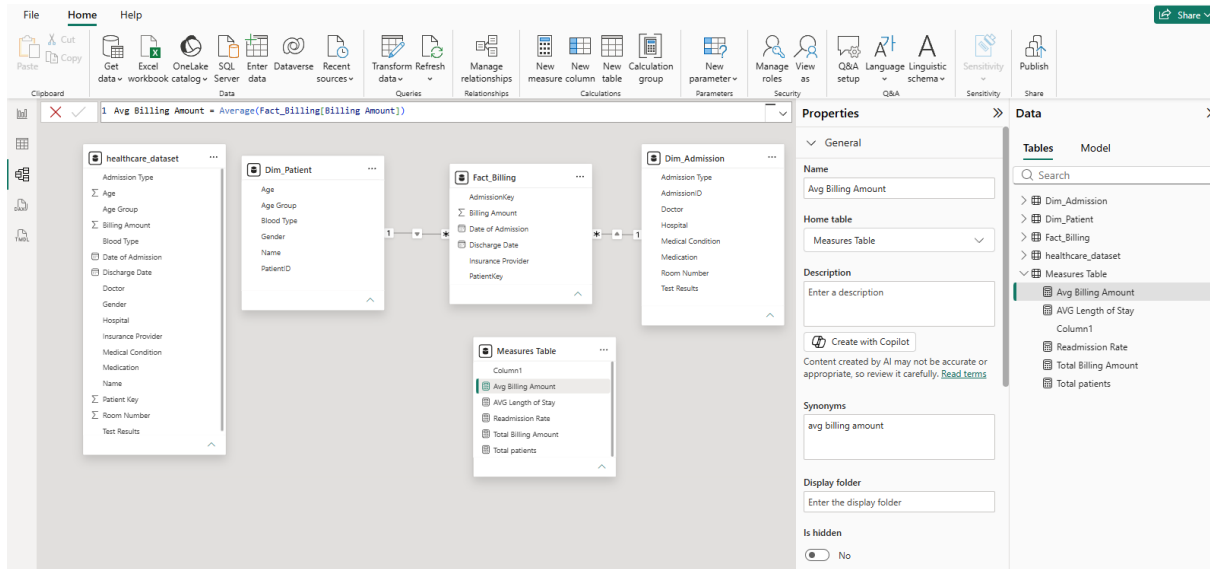
The Measures Table was created to keep the DAX measures separately from the other tables. The measures were created based on the Fact_Billing data and were used for the dashboard.

Measures Table



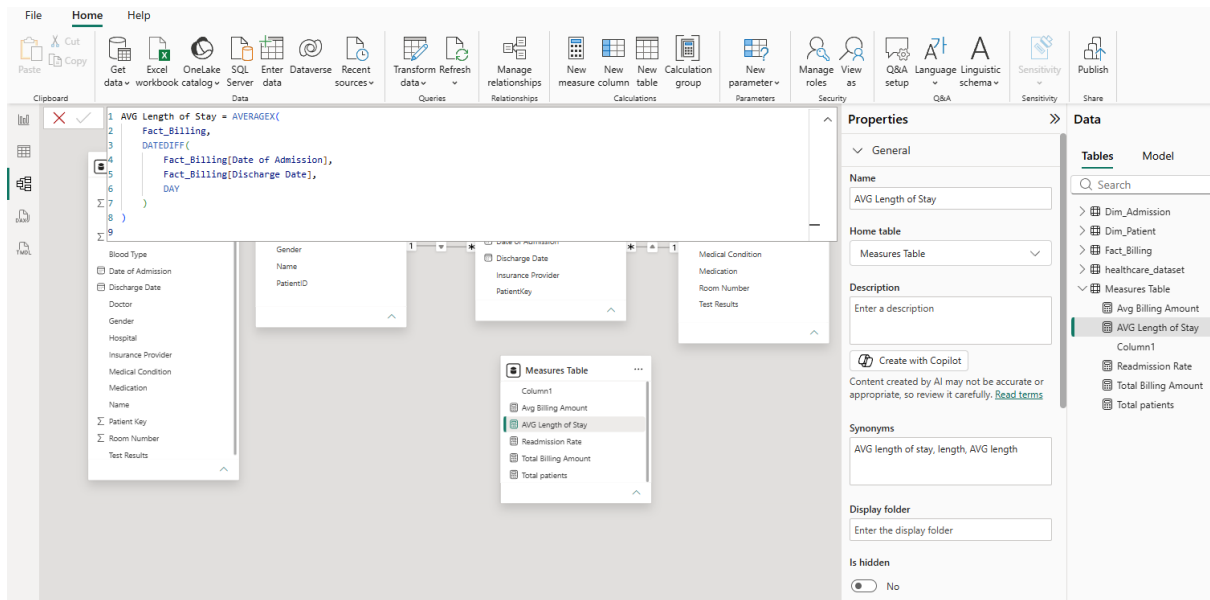
2. Average Billing Amount

The AVG Billing Amount measure was created using AVERAGE on the Billing Amount column. It is used to display the average amount billed.



3. Average Length of Stay

The AVG Length of Stay measure calculates the difference between Date of Admission and Discharge Date using DATEDIFF. AVERAGEX is then used to obtain the average stay duration.



4. Readmission Rate

The Readmission Rate measure was created by identifying patients with more than one admission and comparing the repeat patients with the total patient count.

The screenshot shows the Power BI Desktop interface. The main area displays the DAX formula for the Readmission Rate measure:

```
1 Readmission Rate =  
2 VAR TotalCount = [Total Patients]  
3 VAR RepeatPatients = COUNTROWS(  
4     FILTER(  
5         SUMMARIZE(  
6             Fact_Billing,  
7             Fact_Billing[PatientKey],  
8             "AdmissionCount", COUNTROWS(Fact_Billing)  
9         ),  
10         [AdmissionCount] > 1  
11     )  
12 )  
13 RETURN DIVIDE(RepeatPatients, TotalCount, 0)
```

The right-hand pane shows the Properties window for the measure. The Name is "Readmission Rate", the Home table is "Measures Table", and the Description is "Enter a description". The Synonyms field contains "readmission rate, rate". The Display folder is "Enter the display folder". The Is hidden checkbox is unchecked.

The Data pane on the right shows the model structure, including tables like Dim_Admission, Dim_Patient, Fact_Billing, healthcare_dataset, and Measures Table. The Measures Table contains measures like Avg Billing Amount, AVG Length of Stay, Column1, Readmission Rate, Total Billing Amount, and Total patients.

5. Total Patients

A Total Patients measure was created with COUNTROWS. It counts the records available in the Fact_Billing table and provides the patient count used in the dashboard.

The screenshot shows the Power BI Desktop interface. The main area displays the DAX formula for the Total Patients measure:

```
1 Total patients = COUNTROWS(Fact_Billing)
```

The right-hand pane shows the Properties window for the measure. The Name is "Total patients", the Home table is "Measures Table", and the Description is "Enter a description". The Synonyms field contains "total patients, patients". The Display folder is "Enter the display folder". The Is hidden checkbox is unchecked.

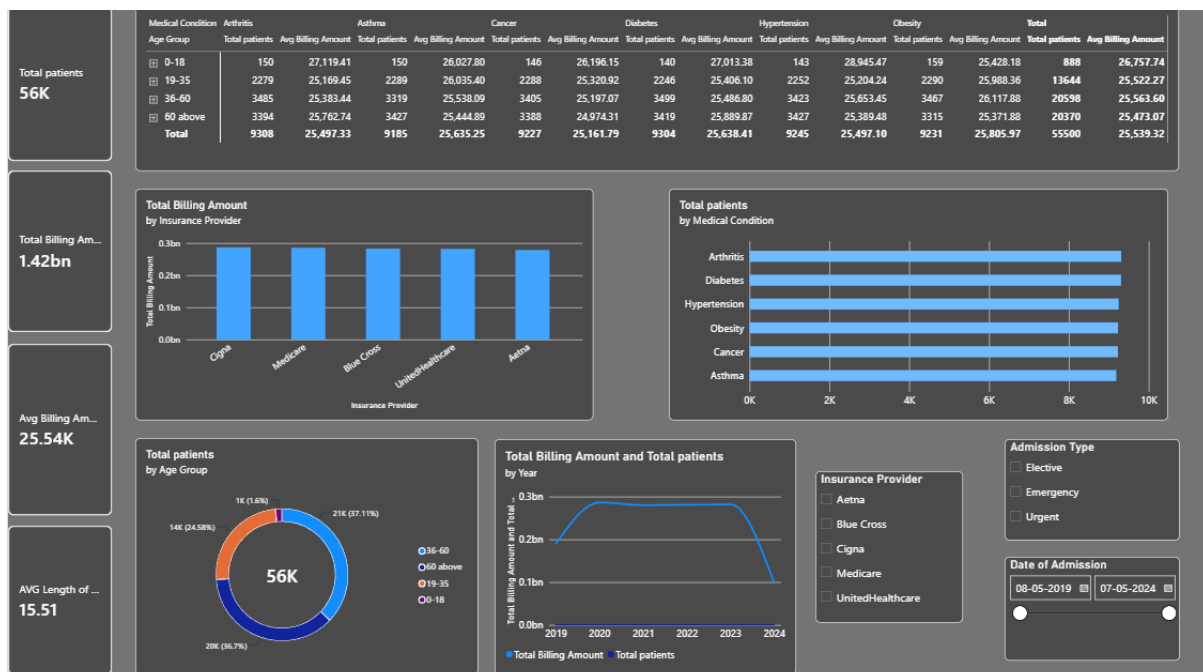
The Data pane on the right shows the model structure, including tables like Dim_Admission, Dim_Patient, Fact_Billing, healthcare_dataset, and Measures Table. The Measures Table contains measures like Avg Billing Amount, AVG Length of Stay, Readmission Rate, Total Billing Amount, and Total patients.

6. Total Billing Amount

The Total Billing Amount measure calculates the overall billing value from the Billing Amount column using SUM.

The screenshot shows the Microsoft Power BI Desktop interface. The main view displays the data model with four tables: **Dim_Patient**, **Fact_Billing**, **Dim_Admission**, and **Measures Table**. The **Fact_Billing** table is connected to **Dim_Patient** and **Dim_Admission**. The **Measures Table** contains a measure named **Total Billing Amount** with the DAX formula: $\text{Total Billing Amount} = \text{SUM}(\text{Fact_Billing}[\text{Billing Amount}])$. The right-hand pane shows the **Properties** for the **Total Billing Amount** measure, including its name, home table, description, and synonyms.

7. Final Dashboard



The final healthcare dashboard includes:

- **KPI Cards** – Total Patients, Total Billing Amount, AVG Billing Amount, and AVG Length of Stay.
- **Matrix** – Patient count and average billing amount by Age Group, Gender, and Medical Condition.
- **Donut Chart** – Patient distribution by Age Group.
- **Stacked Bar Chart** – Patient count by Medical Condition and Admission Type.
- **Line Chart** – Total Billing Amount and Total Patients over time.
- **Clustered Column Chart** – Total Billing Amount by Insurance Provider.
- **Slicers** – Insurance Provider, Admission Type, and Date of Admission for interactive filtering.